



Smell

Identifying a smell out of the many odors in the world around us involves the olfactory system, which isolates different chemicals and then passes signals on to the brain to determine whether they are “good” or “bad.”

What makes a smell?

How we identify smells is still a matter of debate. Research suggests that most odors fall into 10 groups—or primary odors—each of which alerts us to something in the environment. Most smells are made up of a combination of these groups. Smell is a key part of survival, telling us whether something is safe or dangerous.



Fragrant

Light, natural scents such as flowers, grasses, and herbs, typically used in perfumery.



Fruity

Typically includes warm, ripe fruits and other fresh scents that have a sense of smoothness on the nose.



Citrus

Separate from other fruits, citrus has fresh, clean, acidic aromas with a touch of sweetness.



Woody and resinous

Earthy, natural smells, such as compost, fungi, spices, cedar, pine, and mold.



Chemical

Includes synthetic, medicinal, solvent, and gasoline odors that are easily identifiable.



Sweet

Warm, rich, sugary smells with a touch of creaminess, including chocolate, malt, and vanilla.



Minty

Cool, fresh, and invigorating, epitomized by mint, eucalyptus, and camphor.



Toasted and nutty

Slightly burned and caramelized with warm and fatty overtones, such as popcorn and peanut butter.



Pungent

Often unpleasant smells such as manure or sour milk, also onions, garlic, and pickles.



Decayed

Beyond pungent are the odors of rotting food, sewage, household gas, and other “sickening” substances.

WHY DO SMELLS TRIGGER MEMORIES?

Unlike our other senses, smells bypass the thalamus and go straight to the limbic system. Emotions and memories are processed and stored here, especially in the amygdala.

SMELLY OR SWEET?

Dimethyl sulphide (DMS) is a very smelly compound. A whiff of the raw chemical can make you wonder whether something is rotting or if a pungent cheese is in the room. However, flavor chemists find it useful in creating all sorts of tastes. It is used in meat, seafood, milk, egg, wine, beer, vegetable, and fruit flavorings, usually at minuscule concentrations.

